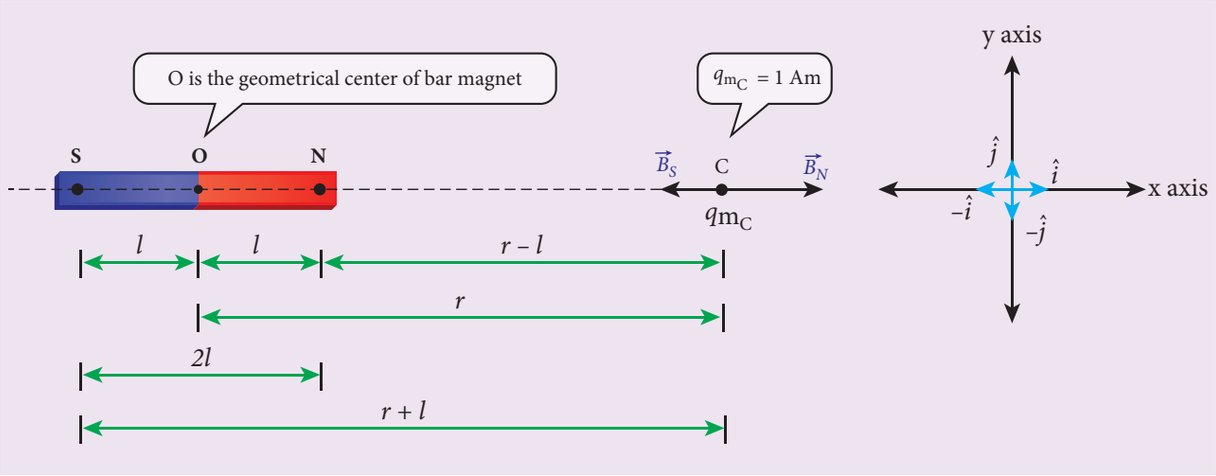


Figure 3.13 Magnetic field at a point along the axial line due to magnetic dipole

at a distance r from the geometrical centre O of the bar magnet can be computed by keeping unit north pole (q_{m_C} = 1 Am) at C.

The magnetic field at C due to the north pole is

$$\vec{B}_N = \frac{\mu_\circ}{4\pi}\frac{q_m}{(r-l)^2}\hat{i} \qquad (3.9)$$

where $(r - l)$ is the distance between north pole of the bar magnet and unit north pole at C.

The magnetic field at C due to the south pole is

$$\vec{B}_S = -\frac{\mu_\circ}{4\pi}\frac{q_m}{(r+l)^2}\hat{i} \qquad (3.10)$$

where $(r + l)$ is the distance between south pole of the bar magnet and unit north pole at C.

The net magnetic field due to the magnetic dipole at point C

$$\vec{B} = \vec{B}_N + \vec{B}_S$$

$$\vec{B} = \frac{\mu_\circ}{4\pi}\frac{q_m}{(r-l)^2}\hat{i} + \left(-\frac{\mu_\circ}{4\pi}\frac{q_m}{(r+l)^2}\hat{i}\right)$$

$$\vec{B} = \frac{\mu_\circ q_m}{4\pi}\left(\frac{1}{(r-l)^2} - \frac{1}{(r+l)^2}\right)\hat{i}$$

$$\vec{B} = \frac{\mu_\circ 2r}{4\pi}\left(\frac{q_m\cdot(2l)}{\left(r^2-l^2\right)^2}\right)\hat{i} \qquad (3.11)$$

Since the magnitude of magnetic dipole moment is $|\vec{p}_m| = p_m = q_m \cdot 2l$, the magnetic field at a point C can be written as

$$\vec{B}_{axial} = \frac{\mu_\circ}{4\pi}\left(\frac{2rp_m}{\left(r^2-l^2\right)^2}\right)\hat{i} \qquad (3.12)$$

If the distance between two poles in a bar magnet is small (looks like short magnet) when compared to the distance between geometrical centre O of bar magnet and the location of point C ($r >> l$),

$$(r^2-l^2)^2 \approx r^4 \qquad (3.13)$$

Therefore, using equation (3.13) in equation (3.12), we get

$$\vec{B}_{axial} = \frac{\mu_\circ}{4\pi}\left(\frac{2p_m}{r^3}\right)\hat{i} = \frac{\mu_\circ}{4\pi}\frac{2}{r^3}\vec{p}_m \qquad (3.14)$$

where $\vec{p}_m = p_m\hat{i}$.

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